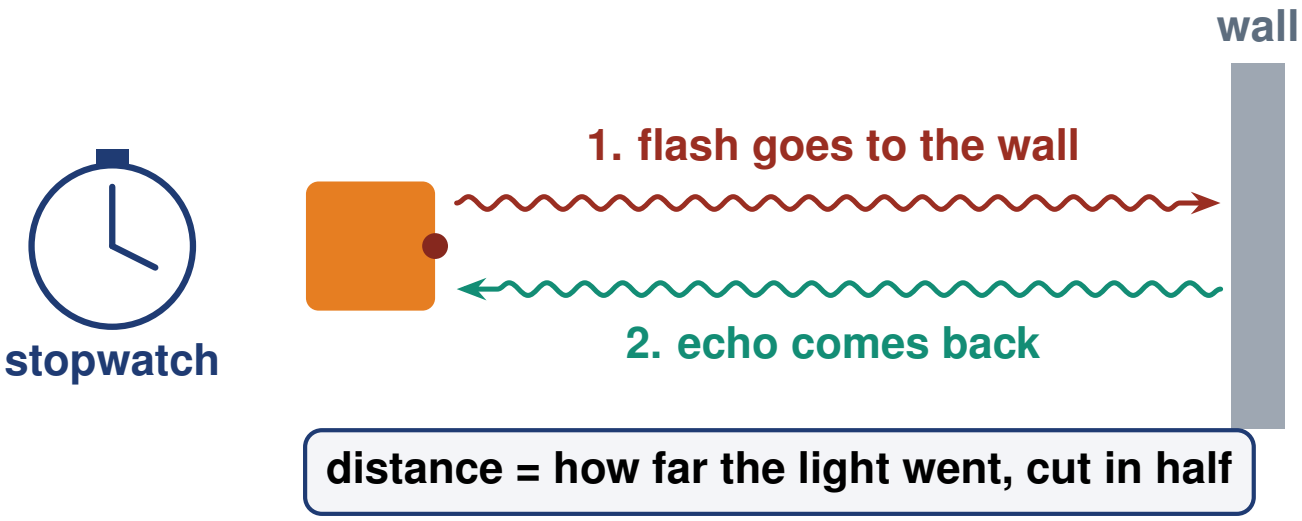
Measuring Distance with Light

BIG IDEA

Time the light there and back, then cut the trip in half.



distance = how far the light went, cut in half

THE IDEA

Light is the fastest thing there is. The LiDAR starts a stopwatch when it flashes and stops it when the echo comes back. The light went *to* the wall and *back*, so the wall is only half as far as the light traveled. That is why we cut the answer in half.

REMEMBER

- ✓ Light travels about 30 cm (one foot) in a billionth of a second.
- ✓ The trip is there *and* back, so we divide by two.
- ✓ If the next flash leaves before the last echo comes home, the LiDAR can mix up which echo is which.

TRY THIS

Shout “Hello!” toward a far wall or a canyon and count until you hear the echo. If you shout again before the echo comes back, can you tell which “Hello” you are hearing?